

4.3 How the derivative affects the shape of the graph (Pages 293 – 304 in the text book)

PART 1:

Increasing / Decreasing Test

- (a) If $f'(x) > 0$ on an interval, then f is increasing on that interval
- (b) If $f'(x) < 0$ on an interval, then f is decreasing in that interval

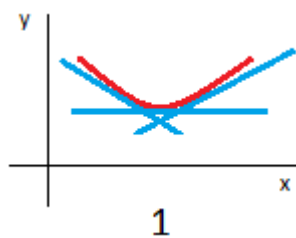
First Derivative Test

If $f'(x)$ changes from positive to negative at c then f has a local maximum at c .
If $f'(x)$ changes from negative to positive at c then f has a local minimum at c .

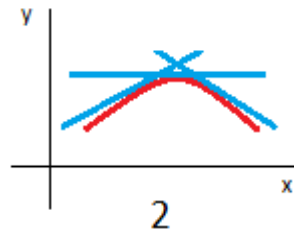
PART 2: Concavity

You can find the definitions of concavity on page 296 of the text book.

A function f is concave upward on an interval I if the graph of f lies above all of its tangents on I . Sketch 1 below shows you what this looks like.



A function f is concave downward on an interval I if the graph of f lies below all of its tangents on I . Sketch 2 below shows you what this looks like.



Concavity test

- (a) If $f''(x) > 0$ on an interval I then f is concave upward on I
- (b) If $f''(x) < 0$ on an interval I then f is concave downward on I

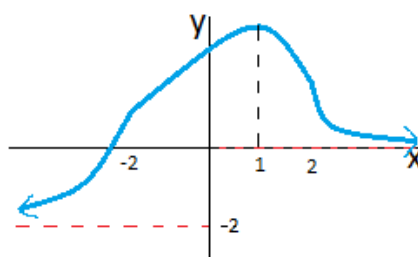
Example 1: (Example 5 in the text book)

Sketch the graph of a function f that has the following properties:

- (1) $f'(x) > 0$ on $(-\infty, 1)$ and $f'(x) < 0$ on $(1, \infty)$
- (2) $f''(x) > 0$ on $(-\infty, -2)$ and on $(2, \infty)$; $f''(x) < 0$ on $(-2, 2)$
- (3) $\lim_{x \rightarrow -\infty} f(x) = -2$ and $\lim_{x \rightarrow \infty} f(x) = 0$

Using the knowledge we have gained thus far, we interpret the properties as follows:

According to (1) the function is increasing on $(-\infty, 1)$ and decreasing on $(1, \infty)$; according to (2) the function is concave upward on $(-\infty, -2)$ and on $(2, \infty)$; according to (3) the curve $y = f(x)$ has horizontal asymptotes at $y = -2$ and $y = 0$. If we put all of this together we find the following sketch:



Inflection point:

A point P on a curve $y = f(x)$ is an inflection point of f if the concavity of f changes at that point.

In the example above, the curve $y = f(x)$ has inflection points at $x = -2$ and $x = 2$.

Example 2:

Find the inflection points of the function f with $f(x) = 1 + 3x + 3x^2 + x^3$.

Solution: $f(x) = 1 + 3x + 3x^2 + x^3$

$$\therefore f'(x) = 3 + 6x + 3x^2$$

$$\therefore f''(x) = 6 + 6x$$

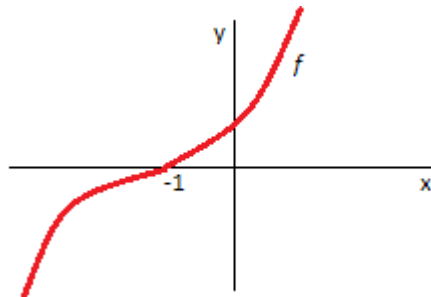
Note that $6 + 6x > 0 \Rightarrow x > -1$ and $6 + 6x < 0 \Rightarrow x < -1$.

Thus, $f''(x) > 0$ if $x > -1$ and $f''(x) < 0$ if $x < -1$.

So f is concave upward on $(-1, \infty)$ and concave downward on $(-\infty, -1)$.

Since the concavity changes at $x = -1$ it follows that $y = f(x)$ has an inflection point at $x = -1$.

The inflection point is the point $(-1, f(-1)) = (-1, 0)$



Second derivative test

Suppose f'' is continuous near c . Then the following hold:

- (a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c
- (b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c

Note: If $f'(c) = 0$ and $f''(c) = 0$ the second derivative does not give a conclusion.

Example:

Use the Second Derivative Test to find the local extremes of the function f with $f(x) = (-x)e^{-3x}$.

Solution:

$$f'(x) = (-1)e^{-3x} + (-x)e^{-3x}(-3) = 3xe^{-3x} - e^{-3x}$$

$$\text{Critical numbers: } f'(x) = 0 \Rightarrow e^{-3x}(3x - 1) = 0$$

Since $e^{-3x} \neq 0$ for all $x \in \mathbb{R}$, it follows from $e^{-3x}(3x - 1) = 0$ that $x = 1/3$ and since $f'(x)$ exists everywhere, $x = 1/3$ is the only critical number.

$$f''(x) = 3e^{-3x} + e^{-3x}(-3)(3x - 1) = 3e^{-3x}(2 - 3x)$$

$$\text{Thus, } f''(1/3) = 3e^{-1}(2 - 1) = 3/e > 0$$

It now follows from the Second Derivative Test that f has a local minimum at $x = 1/3$ and the local minimum is $f(1/3) = (-1/3)e^{-1} = -1/3e$